Research Highlights

Manuscript Title: Risk-Controlled Spatio-Temporal Graph Learning for Anti-Money Laundering Under Extreme Imbalance and Topological Camouflage

Target Journal: Expert Systems with Applications (Elsevier)

- Unified C-STGB framework for AML under extreme class imbalance (<0.05%).
- Continuous tri-band temporal attention prunes 65.9% of camouflage edges.
- Class-conditional conformal risk control guarantees >= 99.0% coverage.
- Outperforms deep GNNs by +43.5 pp macro F1 across 14 financial networks.
- Fast-Path streaming inference achieves 0.45 ms single-event audit latency.

Compliance verification: Exactly five bullet points are provided; each bullet point contains <= 85 characters including spaces, in accordance with the official Elsevier Guide for Authors.